



UNIT – 3

Cloud for IoT

Cloud Computing

- Cloud computing is the on-demand availability of computing resources such as storage space, servers, database, software, networking, analytics etc over the internet.
- Cloud is used as data storage for IoT applications.
- Some of the popular Cloud computing providers are:
- Amazon web service(AWS), Microsoft Azure, Adafruit, ThingSpeak

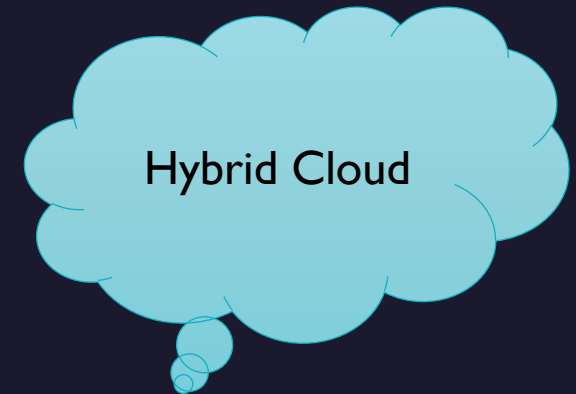
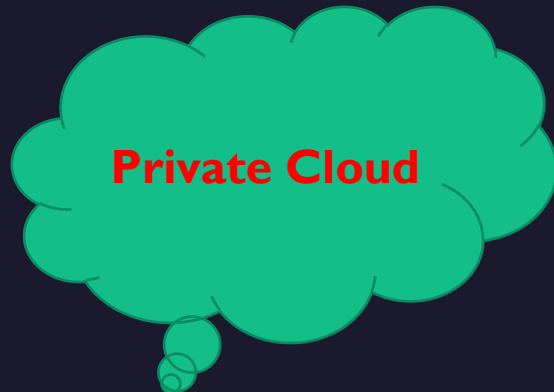
Cloud Deployment Model

Based on the data we work on, cloud are categorized into

1. Private

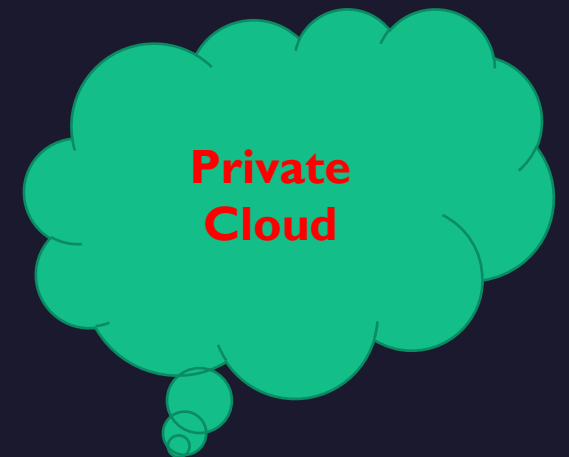
2. Public

3. Hybrid



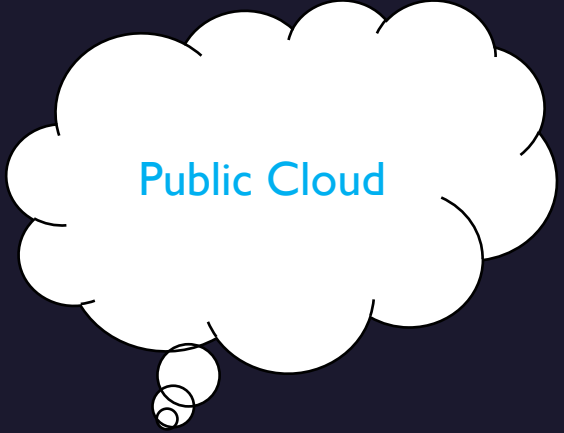
Private Cloud

- In this method, cloud servers are installed inside the organization.
- The hardware, software and infrastructure to be maintained by the organization itself.
- Advantages:
 - Highest data privacy and security.
- Disadvantages:
 - More expensive.
 - Maintenance and upgradation will be difficult.



Public Cloud

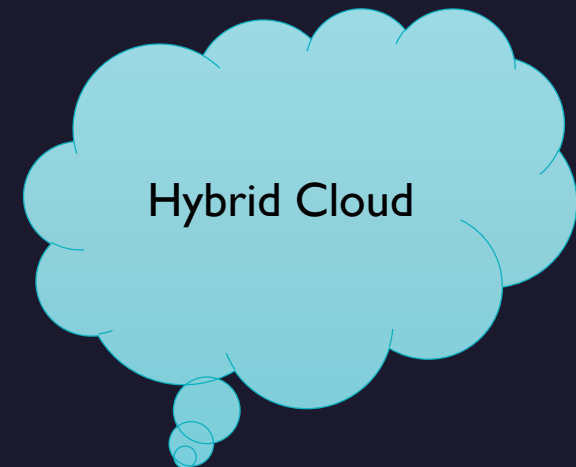
- In this method, cloud services are provided by other third parties.
- The cloud service provider owns the hardware, software and infrastructure
- Advantages:
 - Simple to use
 - Pay and use service
 - 24 x 7 customer support
- Disadvantages:
 - Data not stored in the organization.
 - Security and privacy problems.



Public Cloud

Hybrid Cloud

- It is a combination of public and private cloud deployment.
- Part of the data is stored in the organization and the other part is stored in public cloud.
- Advantages:
 - More flexible to use
 - More secure than public cloud
- Disadvantages:
 - Data security.



Challenges of IoT with Cloud

- The challenges we are facing while building IoT application with cloud can be categorized into 7 types:
 1. Privacy and Security.
 2. Bandwidth Cost.
 3. Migration and Portability.
 4. Reliability and Availability.
 5. Costing.
 6. Data Ownership.
 7. Expertise.

Challenges of IoT with Cloud

I. Privacy and Security

- Valuable and confidential data goes into the cloud, outside the organization.
- Hackers may have tried to access this confidential data.
- Since the data is stored elsewhere, it may be monitored by someone else.
- **Solution:**
- Periodic monitoring, using secured private cloud if the data is too confidential.
- The regulations are to be read before signing a cloud service provider.



Challenges of IoT with Cloud

2. Bandwidth Cost

- IoT applications handles more data.
- So high bandwidth is required to run IoT applications.
- High bandwidth is more expensive.



Challenges of IoT with Cloud

3. Migration and Portability

- If using cloud for first time, all data should be moved to the cloud.
- How much time it will take and how much easier the migration process will be.
- If the current cloud service is not interested to continue, will it be easy to port data to another service provider or private cloud.
- How much would it cost? And will it effect the work flow.



Challenges of IoT with Cloud

4. Reliability and Availability

- IoT is all about continuous monitoring and reading the data. So, the data generation and storage have to be quick and fast.
- 24 x 7 continuous cloud service availability should be there.
- If it misses critical data, it will affect the reliability



Challenges of IoT with Cloud

5. Costing

- We should consider the cost of the service while subscribing.
- Also should monitor data usage to see if it is within the subscription plan.



Challenges of IoT with Cloud

6. Data Ownership

- The data, which we want to store in the cloud, is ours.
- But, data is in the complete custody of the cloud service provider.
- This is a challenge.



Challenges of IoT with Cloud



7. Expertise

- To use cloud with IoT requires specific skills.
- The cloud platform get updated regularly.
- So the experts have to constantly upgrade themselves.



Selection of Cloud for IoT

The following are the important parameters to considered for selecting a cloud for IoT application:

- 1. Certification and standards.**
- 2. Financial status**
- 3. Business/Technology strength**
- 4. Service level agreements**
- 5. Reporting / Tracking**
- 6. Costing and Billing**
- 7. Maintenance, monitoring and upgrade.**
- 8. Support**
- 9. Security**



1. Certification and standards

- Check if the cloud service provider works with industry standards.
- The service provider has got certification like ISO, IEEE etc.



2. Financial Status

- Does the cloud service provider has sufficient funding to run for a long period



3. Business/Technology strength

- Does the cloud service provider have technical expertise and business skill to sustain.



4. Service Level Agreement

- The SLA provides detailed information about the services being provided.



5. Reporting and tracking

- The service provider should give a detailed performance report.



6. Costing and Billing

- The costing and billing should be transparent and should provide the complete details of the usage.



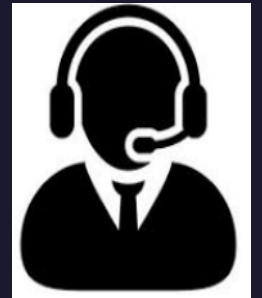
7. Maintenance Monitoring and Upgrade

- It should be easy to install, manage, maintain and upgrade.



8. Support

- Help and assistance should be provided as and when required.



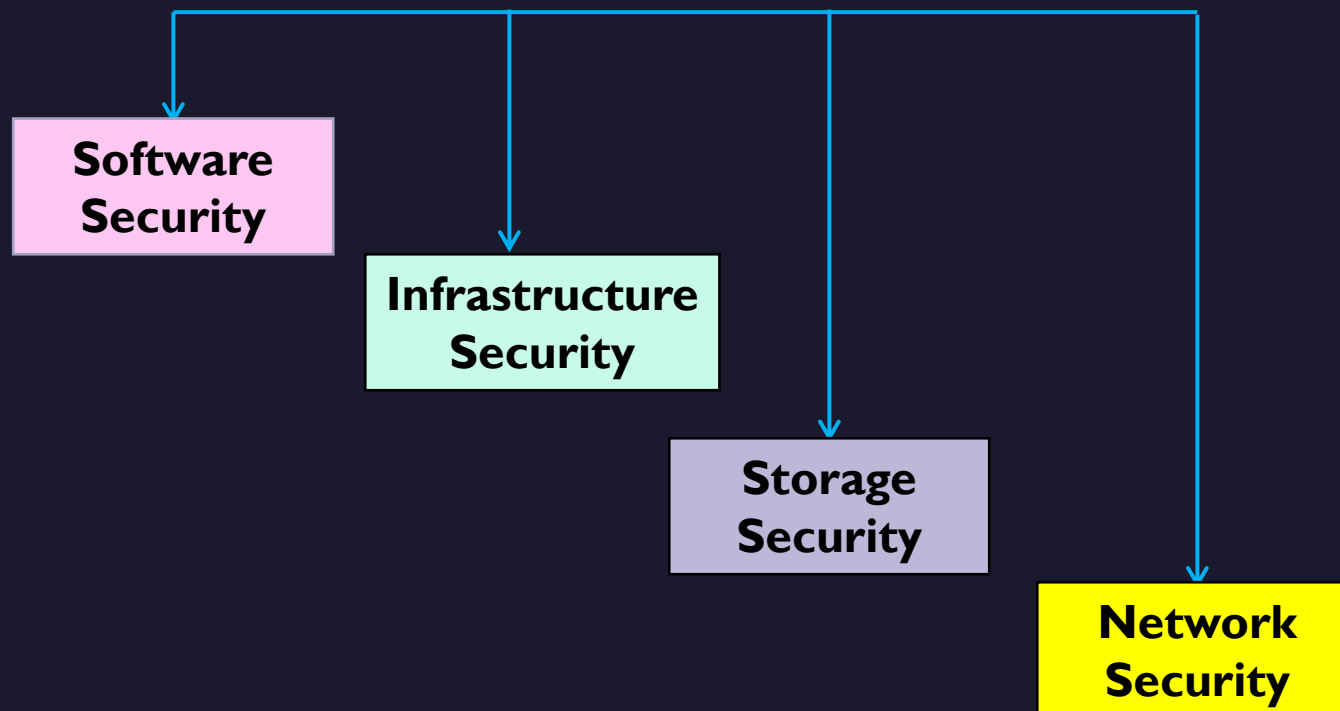
9. Security

- The physical infrastructure and data should be safe and secured.



Cloud Computing – Security aspects

Cloud Computing and Security Aspects can be presented as:



Software Security



- Software plays important role in data security.
- Bugs like memory allocation, design issues, exception handling etc will lead to security issues.



Infrastructure Security



- Make sure that the infrastructure provided by the cloud service provider is safe.
- The entire data stored is depends on this infrastructure.



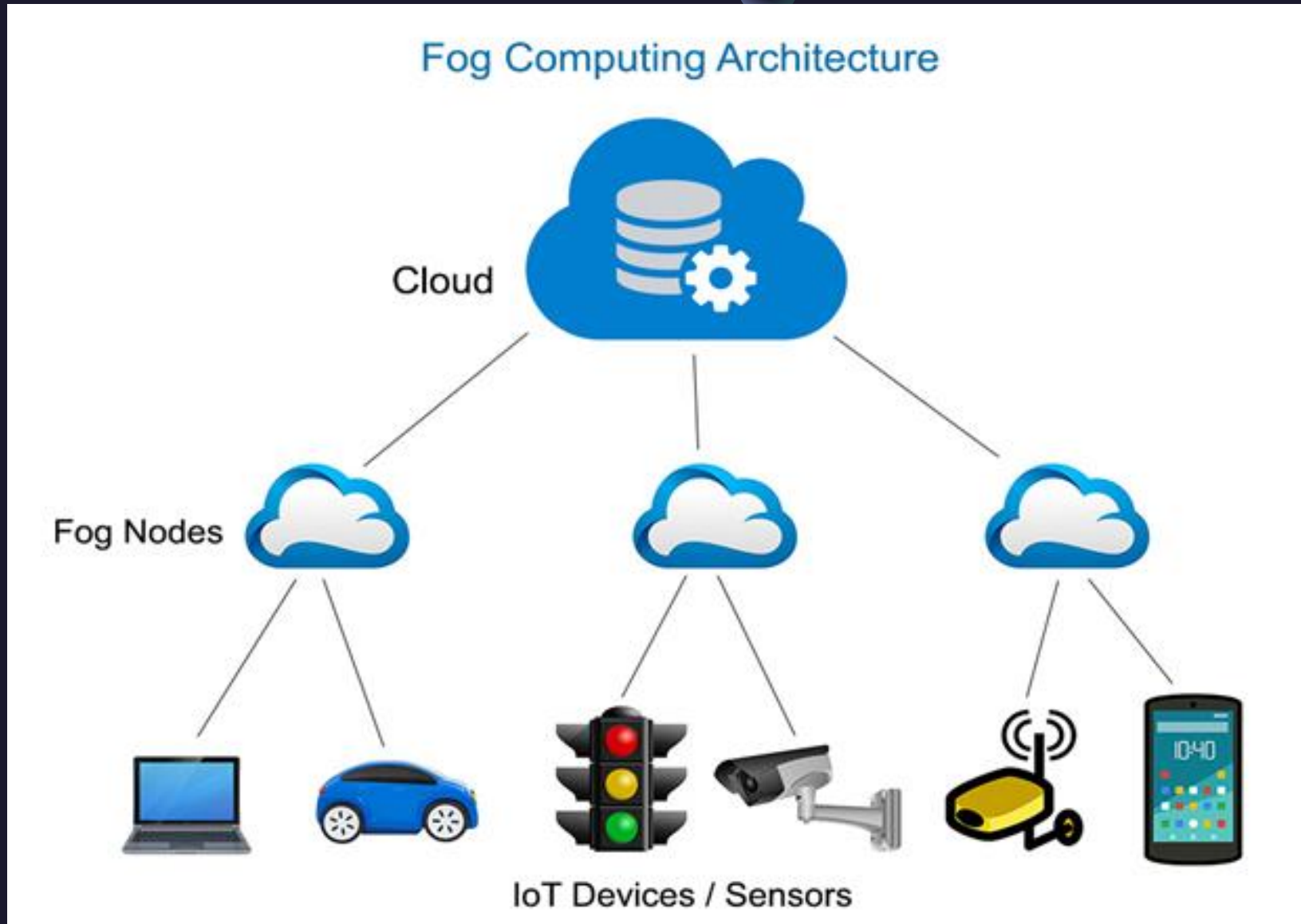
Network Security



- Data moved to and from the cloud through the internet.
- So all attacks happening on the internet will also be relevant for cloud.
- All these threats need to be addressed.



Fog Computing



Fog Computing



- Fog computing is a decentralized computing infrastructure that extends cloud computing capabilities to the edge of the network.
 - It is more closer to the data source and end-users.
 - It was developed to overcome the limitations of traditional cloud computing in certain applications.
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Benefits of fog computing

Since the data is processed closer to the data source, this technology has some benefits over cloud computing.

1. Minimal amount of data send to cloud.
2. Decentralization of data
3. Edge Computing
4. Save bandwidth
5. Reduce data latency(delay)
6. Improve data security
7. Immediate processing of data





Summary